

## Turn In

1. Submit a screenshot of the SSH brute force you performed from Lab 04. (5 points)

```
valid_lft forever preferred_lft forever
santana@l3:~$ echo 'echo "PATH ABUSE"' > ls
santana@l3:~$ chmod +x ls
santana@l3:~$ cp ls whoami
santana@l3:~$ ls
PATH ABUSE
santana@l3:~$ ./rootshell
Verifying you are Preston ...
You are: PATH ABUSE

santana@l3:~$ echo 'echo Preston' > whoami
santana@l3:~$ chmod +x whoami
santana@l3:~$ export PATH=.:$PATH
-bash: export: `PATH=.:usr/local/sbin:usr/local/bin:usr/sbin:usr/bin:sbin:/bin:usr/games:usr/local/games:/snap/bin': not a valid identifier
santana@l3:~$ export PATH=.:$PATH
santana@l3:~$ ./rootshell
Verifying you are Preston ...
You are: Preston

Access Granted
# journalctl -u ssh
Sep 02 12:14:11 l3 systemd[1]: Starting OpenBSD Secure Shell server ...
Sep 02 12:14:11 l3 sshd[1077]: Server listening on 0.0.0.0 port 22.
Sep 02 12:14:11 l3 systemd[1]: Started OpenBSD Secure Shell server.
Sep 02 12:14:11 l3 sshd[1077]: Server listening on :: port 22.
Sep 02 12:44:28 l3 sshd[1077]: Received signal 15; terminating.
Sep 02 12:44:28 l3 systemd[1]: Stopping OpenBSD Secure Shell server ...
Sep 02 12:44:28 l3 systemd[1]: ssh.service: Deactivated successfully.
Sep 02 12:44:28 l3 systemd[1]: Stopped OpenBSD Secure Shell server.
-- Boot 18db80e7db6549529735649b0e293d71 --
Sep 02 12:45:02 l3 systemd[1]: Starting OpenBSD Secure Shell server ...
Sep 02 12:45:02 l3 sshd[1017]: Server listening on 0.0.0.0 port 22.
Sep 02 12:45:02 l3 sshd[1017]: Server listening on :: port 22.
Sep 02 12:45:02 l3 systemd[1]: Started OpenBSD Secure Shell server.
Sep 02 12:49:30 l3 systemd[1]: Stopping OpenBSD Secure Shell server ...
Sep 02 12:49:30 l3 systemd[1]: ssh.service: Deactivated successfully.
Sep 02 12:49:30 l3 systemd[1]: Stopped OpenBSD Secure Shell server.
Sep 02 12:49:30 l3 systemd[1]: Starting OpenBSD Secure Shell server ...
Sep 02 12:49:30 l3 sshd[38706]: Server listening on 0.0.0.0 port 22.
Sep 02 12:49:30 l3 sshd[38706]: Server listening on :: port 22.
```

- 2. Take a moment and list three potential services/options that could be configured to help mitigate brute-force attacks and why**  
(5 points)

Ans) 3 potential services that could be configured to help mitigate brute force attacks are:

| Options                | Why                                                                                                                              |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| PermitRootLogin        | It basically disables any direct root login over ssh and helps in protecting the most privileged account.                        |
| PasswordAuthentication | It basically disables any password authentication, so it prevents guessing of any user's password.                               |
| Max Auth Tries         | It limits the number of authentication attempts per connection, which helps in slowing down the rate of the brute force attacks. |

- 3. Explain two additional methods we can use to prevent brute force attacks. Be specific.**  
(5 points)

Ans) 3 methods that we can use to prevent brute force attacks:

- 1) We can restrict the ssh access using the firewall or VPN by allowing to SSH only from the trusted IP address, or basically allow to SSH when connected through a VPN. It helps the users if the attackers cannot even reach the ssh service, they cannot attempt a brute force attack at all since it effectively hides from the internet.
- 2) We can change the default ssh port since most brute force bots automatically scan and attack and could attack port 22, so if SSH is moved to another port, then these won't usually detect it.
- 3) We can also use Multifactor authentication for SSH as even if the attacker is able to guess steal our password somehow they cannot log in without passing the second authentication, making the brut eforce attack nearly impossible to succeed.
- 4) Fail2ban: It basically monitors authentication firewalls and add temporary firewall rules to block IPs that show repeated failures. It does that because it stops any

automated scanners/bots by cutting off the source IP after a certain number of attempts, by reducing the volume and the success rate of the brute force tries, plus it is easy to tune and protect any services like ssh, ftp, and http by finding out a pattern that matches the logs.

**4. What does Event ID 4672 mean? What privileges are assigned to somebody who logs in under this action? Be specific, no less than five sentences.**

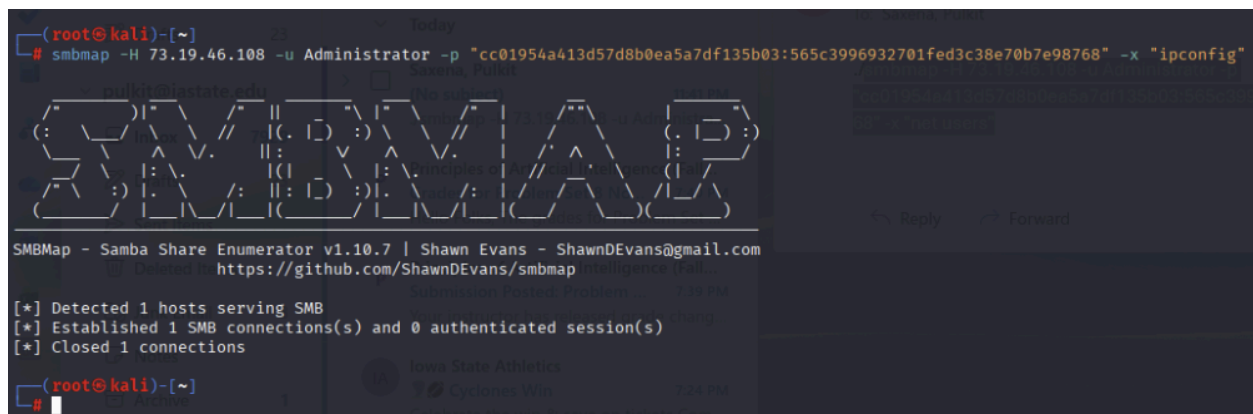
(5 points)

Ans) It basically indicates a new logon that was granted any privileged rights. It also lists any elevated privileges that are assigned, such as SeDebugPrivilege, SeBackupPrivilege, SeTakeOwnershipPrivilege etc., which are more powerful than a normal user's rights and allow the attacker to perform some sensitive tasks such as debugging system processes, it can also act as a part of the OS and could manage the system tokens as well, which may not be noticed by any standard security measures on any system.

Over here, the event record basically keeps a track of all the events such a user account, logon ID and a full list of privileged rights that are assigned. Monitoring this event ID is very important since because it helps the system administrators recognize when privilege access is being granted and whether it has been granted legitimately or not. The security analyst basically use these logs to correlate any kind of privileged logon with other events which helps us in determining any suspicious activity and it also helps in analyzing any incident. The Privileged logons also indicate any misuse of elevated any elevated permissions as attackers can perform privilege escalation either vertically or horizontally.

**5. Screenshot of the failed Pass the Hash attempt in kali.**

(5 points)



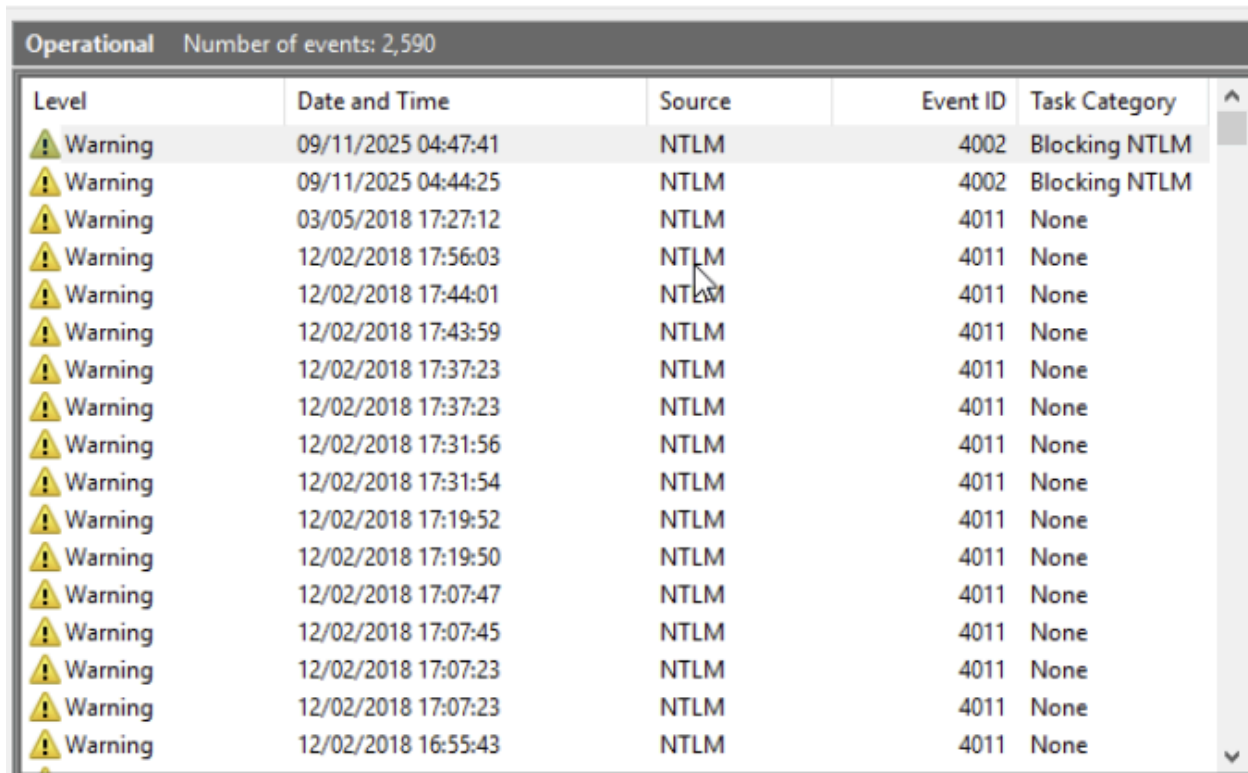
```
(root@kali)-[~]
# smbmap -H 73.19.46.108 -u Administrator -p "cc01954a413d57d8b0ea5a7df135b03:565c3996932701fed3c38e70b7e98768" -x "ipconfig"

SMBMap - Samba Share Enumerator v1.10.7 | Shawn Evans - ShawnDEvans@gmail.com
GitHub: https://github.com/ShawnDEvans/smbmap

[*] Detected 1 hosts serving SMB
[*] Established 1 SMB connections(s) and 0 authenticated session(s)
[*] Closed 1 connections

(root@kali)-[~]
#
```

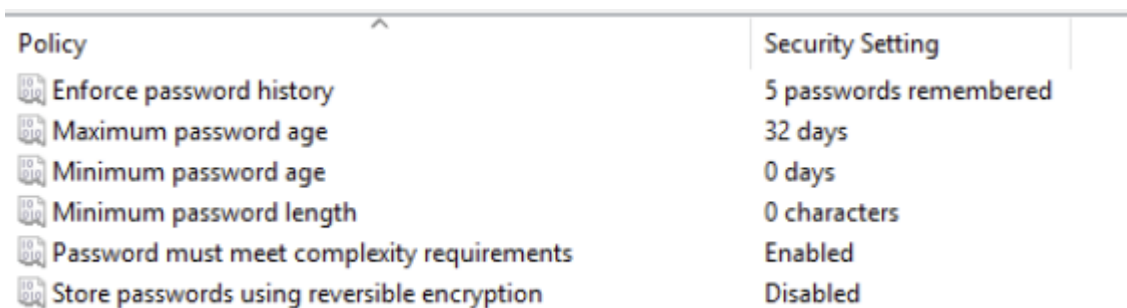
6. Take a screenshot of the failed NTLM login in Windows Event Viewer. What is the Event ID (four-digit code) of the failed login (5 points)



| Operational Number of events: 2,590 |                     |        |          |               |
|-------------------------------------|---------------------|--------|----------|---------------|
| Level                               | Date and Time       | Source | Event ID | Task Category |
| Warning                             | 09/11/2025 04:47:41 | NTLM   | 4002     | Blocking NTLM |
| Warning                             | 09/11/2025 04:44:25 | NTLM   | 4002     | Blocking NTLM |
| Warning                             | 03/05/2018 17:27:12 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:56:03 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:44:01 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:43:59 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:37:23 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:37:23 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:31:56 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:31:54 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:19:52 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:19:50 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:07:47 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:07:45 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:07:23 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 17:07:23 | NTLM   | 4011     | None          |
| Warning                             | 12/02/2018 16:55:43 | NTLM   | 4011     | None          |

The event id is 4002.

7. Screenshot of modified password policy (5 points)



| Policy                                      | Security Setting       |
|---------------------------------------------|------------------------|
| Enforce password history                    | 5 passwords remembered |
| Maximum password age                        | 32 days                |
| Minimum password age                        | 0 days                 |
| Minimum password length                     | 0 characters           |
| Password must meet complexity requirements  | Enabled                |
| Store passwords using reversible encryption | Disabled               |

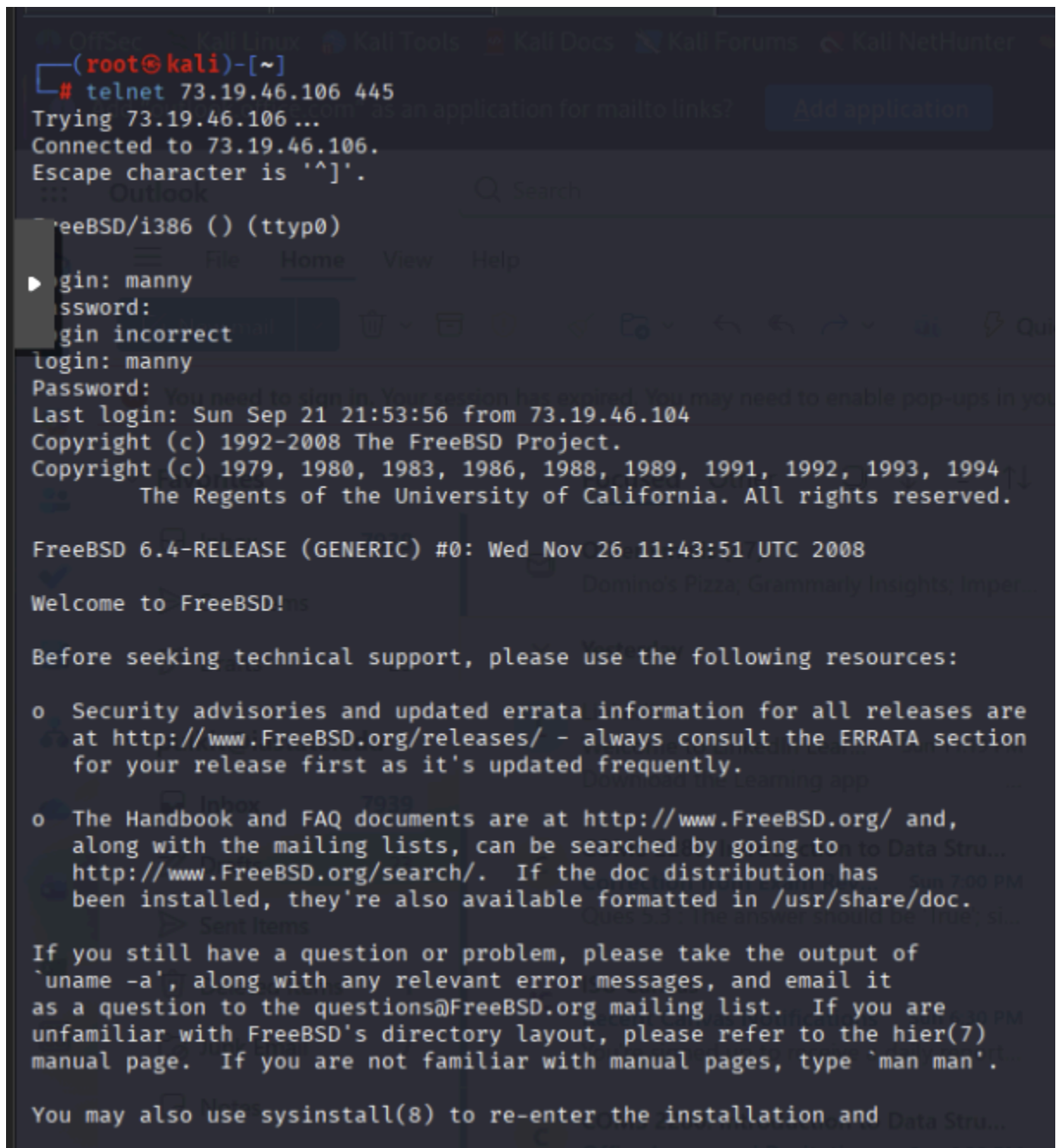
8. What are the “complexity requirements” that we have enforced? (5 points)

Ans) When the complexity requirements are enabled in Windows, then the system basically enforces a mix between the characters and some name exclusion rules as well. So, basically a

password must contain Uppercase and lowercase letters, digits and any special characters like !, #, @ etc. Secondly, it must not contain the user's account name or even parts of his full name.

## 9. Screenshot of successful telnet login

(5 points)



```
(root@kali)-[~]
# telnet 73.19.46.106 445
Trying 73.19.46.106 ...
Connected to 73.19.46.106.
Escape character is '^]'.
FreeBSD/i386 () (ttyp0)
login: manny
password:
login incorrect
login: manny
Password:
Last login: Sun Sep 21 21:53:56 from 73.19.46.104
Copyright (c) 1992-2008 The FreeBSD Project.
Copyright (c) 1979, 1980, 1983, 1986, 1988, 1989, 1991, 1992, 1993, 1994
The Regents of the University of California. All rights reserved.

FreeBSD 6.4-RELEASE (GENERIC) #0: Wed Nov 26 11:43:51 UTC 2008

Welcome to FreeBSD!

Before seeking technical support, please use the following resources:

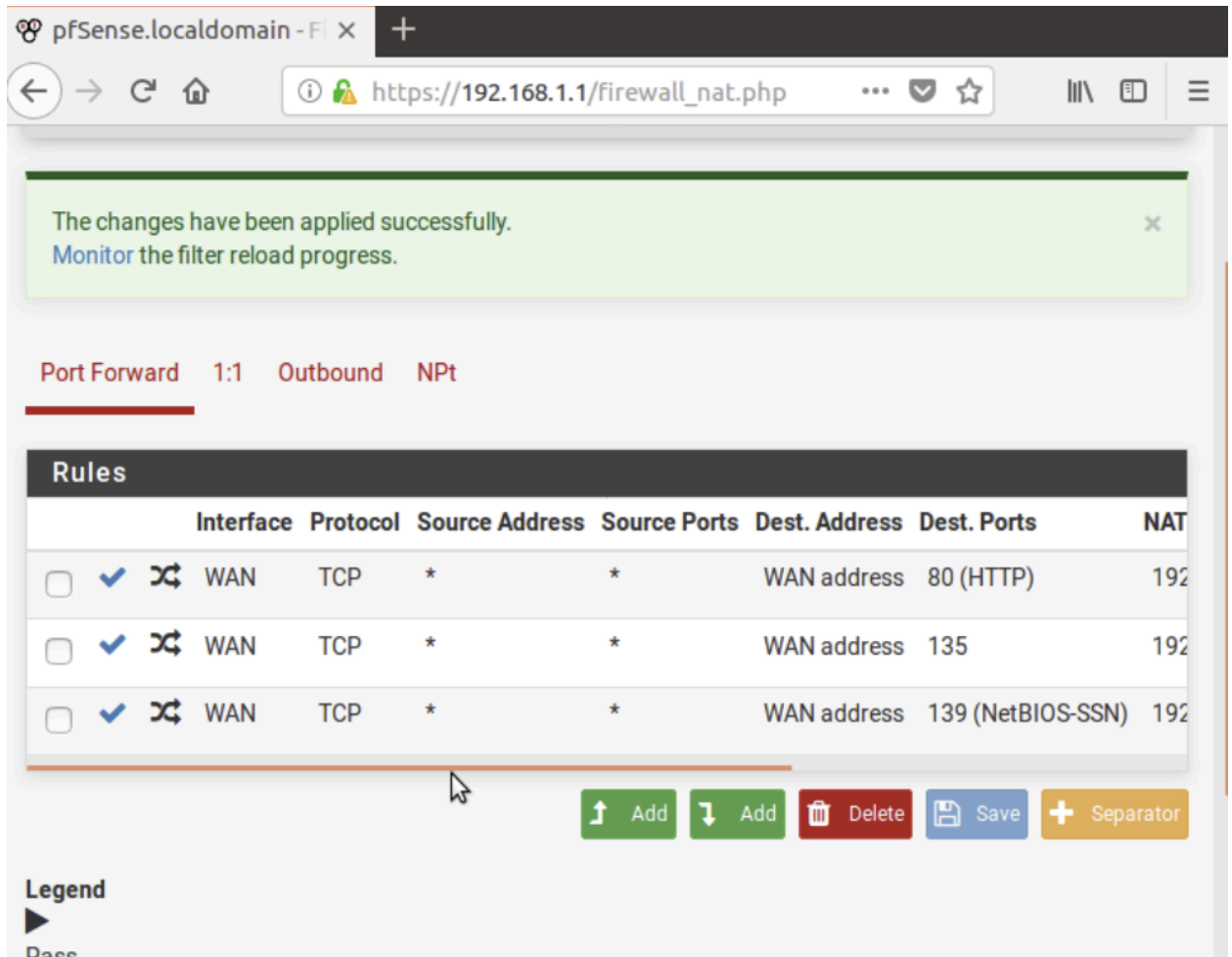
o Security advisories and updated errata information for all releases are
  at http://www.FreeBSD.org/releases/ - always consult the ERRATA section
  for your release first as it's updated frequently.

o The Handbook and FAQ documents are at http://www.FreeBSD.org/ and,
  along with the mailing lists, can be searched by going to
  http://www.FreeBSD.org/search/. If the doc distribution has
  been installed, they're also available formatted in /usr/share/doc.

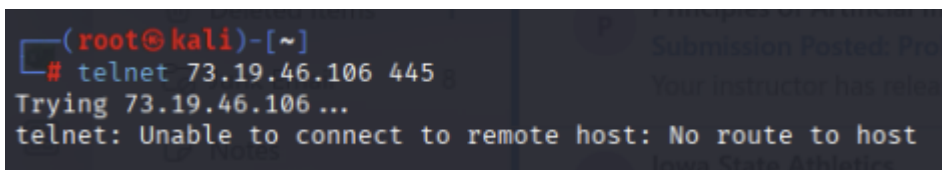
If you still have a question or problem, please take the output of
`uname -a`, along with any relevant error messages, and email it
as a question to the questions@FreeBSD.org mailing list. If you are
unfamiliar with FreeBSD's directory layout, please refer to the hier(7)
manual page. If you are not familiar with manual pages, type `man man`.

You may also use sysinstall(8) to re-enter the installation and
```

10. Screenshot of updated pfSense port-forwarding rules  
(5 points)



11. Screenshot of failed telnet login  
(5 points)



**12. Screenshot of listening ports on XX.XX.XX.104**

(5 points)

```
cpre2310@ISEage:~$ netstat -nat
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:80               0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:22               0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:443              0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:3306              0.0.0.0:*               LISTEN
tcp6       0      0 :::22                   :::*                     LISTEN
cpre2310@ISEage:~$
```

**13. What service is running at port 3306?**

(5 points)

Ans) MySQL runs on port 3306.

**14. Screenshot of default incoming/outgoing UFW policies**

(5 points)

```
cpre2310@ISEage:~$ sudo ufw enable
Firewall is active and enabled on system startup
cpre2310@ISEage:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing)
New profiles: skip
```



15. Screenshot of newly created UFW rules  
(5 points)

```
cpre2310@ISEage:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
cpre2310@ISEage:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
cpre2310@ISEage:~$ sudo ufw allow 443/tcp
Rule added
Rule added (v6)
cpre2310@ISEage:~$ sudo ufw status
Status: active
```

| To      | Action | From          |
|---------|--------|---------------|
| ---     | -----  | ----          |
| 22/tcp  | ALLOW  | Anywhere      |
| 80/tcp  | ALLOW  | Anywhere      |
| 443/tcp | ALLOW  | Anywhere      |
| 22/tcp  | ALLOW  | Anywhere (v6) |
| 80/tcp  | ALLOW  | Anywhere (v6) |
| 443/tcp | ALLOW  | Anywhere (v6) |

```
cpre2310@ISEage:~$
```



**16. Screenshot of nmap scan results showing blocked port 3306**

(5 points)

```
(root@kali)-[~]
# nmap -p 3306 73.19.46.104
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 15:59 CST
Nmap scan report for 73.19.46.104
Host is up (0.00056s latency).

PORT      STATE SERVICE
3306/tcp  filtered mysql
MAC Address: 00:02:31:00:6C:04 (Ingersoll-Rand)

Nmap done: 1 IP address (1 host up) scanned in 13.46 seconds

(root@kali)-[~]
#
```

**17. Specify which domain these packages are being downloaded from. Why?**

(5 points)

Ans) This is the domain: <https://kernel.ubuntu.com/~kernel-ppa/mainline/v3.16.45/>

**18. Screenshot of upgraded kernel version**

(5 points)

```
Ubuntu 13.10 ISEage tty1

ISEage login: cpre2310
Password:
Last login: Sun Nov  9 04:31:39 CST 2025 on tty1
cpre2310@ISEage:~$ uname -a
Linux ISEage 3.16.45-031645-generic #201707030336 SMP Mon Jul 3 07:40:31 UTC 2017 x86_64 x86_64 x86_64 GNU/Linux
cpre2310@ISEage:~$
```

19. Screenshot of “hung” Dirty Cow  
(5 points)

```
Ubuntu 13.10 ISEage tty1
ISEage login: scrat
Password:
Last login: Sun Nov  9 04:02:08 CST 2025 on tty1
scrat@ISEage:~$ ./cow
DirtyCow root privilege escalation
Backing up /usr/bin/passwd to /tmp/bak
Size of binary: 47032
Racing, this may take a while..
thread stopped
thread stopped
```

**20. Identify each version you would have to upgrade through to get from 13.10 to the current LTS**

(5 points)

Ans) In order to get an update from Ubuntu 13.10 to the current LTS, we would have to upgrade through the LTS releases as the upgrades are supported from one release to the next one in sequence, so the upgrade path that we must follow is:

13.10 → 14.4 LTS → 16.04 LTS → 18.04 LTS → 20.04 LTS → 22.04 LTS → 24.04 LTS